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Title:	**TOPIC ABSTRACT FORM**
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Program	Bachelor of Science in Computer Science

Proposed Title: **Heterogeneous Spatiotemporal Graph Neural Network for AIS–SAR Vessel Association and Dark-Vessel Candidate Detection**

Background of the Proposed Topic

Maritime transportation generates substantial vessel activity that requires continuous monitoring across the maritime domain. As of January 2026, the global merchant fleet comprised approximately 116,000 vessels of at least 100 gross tons (GT), with a total carrying capacity of 2.5 billion deadweight tons (DWT) (UNCTAD, 2026). This operational volume is directly reflected in the Philippines, where coastal waterways and strategic chokepoints support extensive domestic and international shipping. In 2024 alone, Philippine ports documented over 615,000 ship calls at berth, highlighting the critical necessity for robust maritime traffic monitoring (Philippine Statistics Authority [PSA], 2026).

The Automatic Identification System (AIS) is one of the widely used systems for marine monitoring. It supports maritime safety and navigation by transmitting static identification details, such as the vessel's name and Maritime Mobile Service Identity (MMSI), as well as dynamic navigational information, including position, course, speed, and navigational status (Shou & Friis, 2023). When available over time, AIS observations can reconstruct vessel trajectories and analyze maritime traffic and movement patterns (Troupiotis-Kapeliaris et al., 2025).

Despite its widespread utility, AIS does not provide a complete representation of actual vessel presence. AIS data remain susceptible to missing values, abnormal records, communication problems, signal interference, and positional inconsistencies (He et al., 2021). Missing AIS observations may also result from signal blind areas or packet loss during transmission, which can lead to incomplete vessel trajectories (He et al., 2021). Furthermore, during a technical consultation with the Maritime Industry Authority (MARINA) Regional Office IV, it was clarified that AIS carriage requirements apply only to specific categories of vessels under applicable maritime regulations, meaning that some vessels may legitimately operate without AIS transponders. The consultation also identified equipment malfunctions and intentional transponder deactivation as possible causes of absent AIS transmissions. In addition, Welch et al. (2022) identified suspected intentional AIS disabling events that obscure vessel activity in commercial fishing data. These limitations make it difficult to determine whether the absence of an AIS observation represents a temporary data or transmission issue or an actual discrepancy in reported vessel activity.

Synthetic Aperture Radar (SAR) provides a complementary source of vessel observations because it can detect vessels independently of their AIS transmissions (Heiselberg et al., 2023). Recent studies have demonstrated the use of SAR for automated vessel analysis. (Bhattacharjee et al., 2024), for example, developed an attention-guided deep-learning approach for classifying ships in low-resolution Sentinel-1 SAR imagery, demonstrating the applicability of SAR data for automated vessel recognition. Heiselberg et al., (2023) further demonstrated that ship velocity information can be estimated from SAR imagery using multitask deep learning, showing that SAR observations can contain motion-related characteristics useful for relating satellite detections to vessel trajectories.

The combination of AIS and SAR creates an opportunity to examine whether a vessel observed in SAR imagery has a corresponding AIS trajectory. (Li et al., 2025), for example, developed a multi-feature association method that matched remotely sensed vessel detections with AIS data using position, vessel dimensions, and course information. Their findings show the potential of AIS-SAR association while also indicating that correspondence becomes more challenging in dense scenes. Therefore, the use of both data sources does not by itself resolve the problem of determining which AIS trajectory corresponds to a SAR-detected vessel.

A lack of direct correspondence between a SAR-detected vessel and an AIS trajectory, however, does not by itself establish that the vessel is intentionally non-broadcasting. An apparent mismatch may result from

temporary AIS transmission gaps, differences between SAR and AIS observation times, positional inconsistencies, or ambiguity among nearby vessels. Recent work by Chen et al. (2026) further demonstrates that SAR-AIS association can become ambiguous when several AIS candidates are spatially close to a SAR detection, particularly when SAR geolocation error, AIS position prediction error, spatial disturbances, and motion-induced displacement are considered. The study also shows the need to align AIS trajectories with the SAR imaging epoch and to consider multiple vessel characteristics rather than relying solely on nearest-neighbor association Chen et al. (2026). In congested maritime environments, these factors can make it difficult to determine whether an apparent mismatch represents a genuine AIS-SAR discrepancy or an explainable association or data-quality issue. Consequently, an unmatched SAR vessel cannot automatically be treated as a dark-vessel candidate.

Interpreting these cases therefore requires more than comparing the distance between a SAR detection and a single AIS observation. Position, observation time, speed, course, trajectory continuity, and relationships with nearby vessels can provide additional context for assessing whether an AIS trajectory is consistent with a SAR observation. Graph-based approaches are relevant because they can represent vessels and their relationships across space and time. Zhang et al. (2023) developed a gated spatio-temporal graph aggregation network that modeled spatial interactions and temporal dependencies among vessels for trajectory prediction. Wang et al. (2024) similarly represented vessel states as nodes and modeled interactions among neighboring vessels in complex and crowded sea areas using a spatio-temporal graph. These studies demonstrate the use of spatiotemporal graph models to represent spatial interactions and temporal dependencies among vessels.

Contextual information can also help interpret potentially unusual vessel observations. Forti et al. (2022) incorporated AIS, radar, and contextual information to identify deviations from expected vessel behavior, supporting the consideration of surrounding vessel activity and normal maritime behavior when interpreting potential AIS-SAR discrepancies. However, such anomaly-detection approaches primarily focus on behavioral anomaly detection rather than cross-modal vessel correspondence.

Existing studies demonstrate the potential of SAR-based vessel detection, AIS–SAR association, and graph-based vessel modeling. However, accurately matching vessels and distinguishing genuine discrepancies from AIS transmission gaps, temporal or spatial differences, and nearby vessel ambiguity remain challenging. To address this gap, the study proposes a heterogeneous spatiotemporal graph neural network that integrates spatial, temporal, kinematic, and anchorage-aware contextual information to improve vessel association and reduce false-positive dark-vessel candidate indications. Its effectiveness will be evaluated through baseline comparisons and ablation studies.

This problem is particularly relevant to Batangas Bay, where substantial vessel activity and closely positioned vessels can make AIS-SAR correspondence more difficult. In such conditions, temporary AIS gaps, multiple nearby trajectories, and routine vessel behavior may contribute to false-positive discrepancy indications. To address this problem, the study proposes a cross-modal spatiotemporal graph framework that uses AIS trajectories and SAR vessel observations to assess vessel correspondence and identify dark-vessel candidates while considering spatial, temporal, and contextual relationships. The study therefore focuses on the following problems:

- The difficulty of accurately establishing correspondence between SAR-detected vessels and AIS trajectories, particularly when multiple AIS observations may be spatially or temporally associated with a single SAR detection.
- The difficulty of distinguishing genuine AIS-SAR discrepancies from temporary AIS transmission gaps, spatial or temporal mismatches, and ambiguous associations among closely positioned vessels.
- The occurrence of false-positive discrepancy indications caused by normal maritime conditions, including vessel congestion, slow movement, loitering, and other routine vessel interactions.

Objectives of the Study

General Objective:

The primary objective of this study is to formulate and evaluate a cross-modal spatiotemporal graph neural network that associates Sentinel-1 SAR satellite observations with AIS vessel trajectories to detect correspondence discrepancies and identify dark-vessel candidates, while suppressing false-positive anomalies induced by harbor congestion and routine anchorage loitering in Batangas Bay.

Specific Objectives:

1. Curate and synchronize historical Sentinel-1 SAR observations and AIS telemetry for Batangas Bay within a kinematically bounded ± 15-minute window via trajectory spline interpolation, establishing an ambiguity-aware ground-truth labeling protocol under time-blocked and vessel-disjoint partitioning to preclude data leakage.

2. Formulate a dynamic heterogeneous spatiotemporal graph neural network with cross-modal attention and an anchorage-aware contextual filter to evaluate AIS-SAR candidate associations and suppress false-positive discrepancy indications from routine harbor loitering.

3. Benchmark vessel association and discrepancy detection performance against a Classical Hungarian Bipartite Matching baseline and a Pairwise Multi-Layer Perceptron, evaluating False-Positive Rate (FPR), Precision, Recall, and F1-score across harbor traffic density tiers and controlled synthetic AIS dropouts (10% to 90%).

4. Develop a lightweight interactive web-based proof-of-concept (POC) prototype that visualizes SAR detections, matches AIS trajectories, candidate association probabilities, and anchorage geofences to demonstrate automated decision-support outputs.

Significance of the Study

As maritime surveillance in congested chokepoints like Batangas Bay faces operational bottlenecks, this research introduces a unified cross-modal graph framework that integrates satellite SAR detections with cooperative AIS telemetry. The primary computer science contribution of this study lies in its methodological advancement of multimodal data fusion and relational deep learning: it models cross-modal entity association as link prediction over a heterogeneous spatiotemporal graph, while utilizing contextual geofencing to systematically suppress false-positive alerts induced by routine harbor loitering and benign transmission dropouts.

The system provides targeted monitoring support across several sectors. **Maritime Authorities and Coast Guards** benefit from a tool that distinguishes potential anomalies from normal transmission gaps, allowing them to prioritize unexplained AIS-SAR discrepancies for further manual verification. **Port Administrators** receive supplementary data to manage localized vessel traffic more effectively and maintain navigational safety. For **Environmental Protection Agencies**, the framework serves as an improved maritime monitoring support tool that identifies unexplained observations requiring further verification. Furthermore, **Future Researchers** gain access to a curated AIS-SAR dataset and a baseline architecture for multimodal anomaly detection.

The study introduces three distinct technical contributions:

1. Cross-Modal Spatiotemporal Edge Formulation: Designs a dynamic bipartite graph structure that links asynchronous radar detections with spline-interpolated AIS trajectories through continuous kinematic divergence vectors (Haversine distance, speed differential, heading deviation, and observation latency).
2. Anchorage-Aware Contextual Attention**:** Integrates domain-specific harbor geofencing derived from official nautical charts to dynamically mask low-risk, berth-waiting targets, thereby suppressing false alarms in high-density anchorage zones.
3. Objective Evaluation Protocol: Establishes an objective labeling protocol and tests the framework against standard baseline methods using a vessel-disjoint evaluation strategy and controlled synthetic AIS dropout benchmarks.

By focusing on the performance of the AIS-SAR matching and graph-based network, this research delivers a data-driven solution for complex harbor environments. Ultimately, this computational framework aligns with SDG 14: Life Below Water by supporting improved maritime monitoring and identifying vessel observations that may warrant further investigation for potential environmental or regulatory concerns.

Methodology

Research Design and Study Area

This study will employ a quantitative experimental research design to develop and evaluate a cross-modal spatiotemporal graph framework for AIS-SAR vessel association and dark-vessel candidate detection. The study will focus on a defined portion of Batangas Bay, defined by a spatial bounding box bounded by latitudes 13.61° N to 13.83° N and longitudes 120.91° E to 121.08° E Flanders Marine Institute, 2026). This designated area captures the core port activities, adjacent anchorages, and approaching vessel traffic necessary for cross-modal association.

Dataset Sources and Feasibility Evaluation

The study will use SAR vessel observations and AIS trajectory records covering Batangas Bay. Historical observations from 2025 will be considered as the initial study period, subject to the availability of usable data from both sources. The final observation period and dataset size will be determined after assessing geographic coverage, timestamp consistency, completeness, and suitability for vessel association.

Sentinel-1 Ground Range Detected (GRD) imagery will be sought through the Copernicus Data Space Ecosystem. Global Fishing Watch's SAR vessel-detection dataset will also be examined as a potential source of detection locations, acquisition timestamps, estimated vessel lengths, and available association information. These detection records will be distinguished from the original SAR imagery. Available fields will be assessed for their suitability for the proposed framework before inclusion in the dataset. An initial collection of Global Fishing Watch Sentinel-1 SAR vessel detections for Batangas Bay in 2025 has been obtained and filtered, yielding 344 detections across 23 acquisition scenes, including 18 detections provisionally classified as unmatched by GFW.

Access to historical AIS records will be sought through a formal request to the Philippine Ports Authority–Port Management Office of Batangas. The request will cover vessel-position records and, where available, vessel identifiers, timestamps, speed over ground, course over ground, and vessel dimensions. Access to these records remains unconfirmed. Moreover, commercial provider VesselFinder has confirmed the availability of 10,922 historical AIS position records from 404 unique MMSIs across the corresponding SAR acquisition windows. However, these AIS records have not yet been acquired, and their suitability for constructing validated AIS–SAR associations remains to be assessed.

Data feasibility will be assessed by identifying periods with overlapping SAR acquisitions and AIS observations within the study area. The assessment will consider the number of usable scenes, the frequency and continuity of AIS reports, missing fields, and the availability of supporting information for reference labeling. Dataset counts and training, validation, and testing partitions will be established after this assessment. Supplementary port records will support verification where access is granted and relevant evidence is available.

The prototype's operating mode will be determined after data access and coverage have been assessed. If suitable historical AIS and SAR observations are available, vessel association and discrepancy assessment will be demonstrated through the replay of recorded scenes. A live or periodically updated demonstration will be considered only if continuing access to sufficiently timely data can be secured. If overlapping observations are insufficient for the planned model development and evaluation, the observation period or study scope will be reassessed before implementation.

Spatiotemporal Alignment, Reference Labeling, and Ground-Truth Validation

SAR vessel detections will be aligned with available historical AIS trajectories using the SAR acquisition timestamps. An initial ±15-minute window, covering 15 minutes before and after each acquisition, will be used to retrieve AIS observations for candidate association. The suitability of this window for Batangas Bay will be assessed based on AIS reporting frequency, reception gaps, and the availability of observations surrounding each acquisition. Alternative window lengths will be examined through sensitivity analysis.

Where sufficient observations are available, cubic Hermite spline interpolation will be used to estimate vessel positions at the SAR acquisition time. Interpolated positions will be checked for implausible displacement and abrupt changes in motion. Sparse or inconsistent trajectories will be flagged as uncertain rather than treated as reliable position estimates. Candidate associations will be assessed using spatial proximity, course, speed, trajectory continuity, and available vessel characteristics, drawing on multi-feature association approaches such as Li et al. (2025). Incomplete AIS reception will also be considered when interpreting missing observations (Kroodsma et al., 2022).

Ground-truth validation will seek supporting evidence beyond the AIS observations used for candidate association. Where access is granted, vessel traffic monitoring records, radar observations, and port movement records from the Philippine Ports Authority–Port Management Office of Batangas will be examined to verify vessel presence and identity near the SAR acquisition times. Their suitability will depend

on timestamp precision, geographic coverage, and the information recorded. Records derived from the same AIS observations used for association will be identified separately and will not be treated as independent confirmation.

Two reviewers will independently assess the available SAR observations, AIS trajectories, and supporting records to assign reference labels without access to the proposed model's predictions. Each case will be assigned to one of three categories:

- Matched Vessel: Sufficient evidence supports correspondence between a SAR detection and a specific AIS vessel.
- Unresolved Discrepancy Candidate: A credible SAR vessel detection has no reliable AIS correspondence despite sufficient AIS coverage to assess the case, and the available evidence does not adequately explain the discrepancy. This label does not establish intentional AIS disabling.
- Ambiguous/Insufficient Evidence: Competing associations, incomplete AIS coverage, uncertain SAR detections, or missing supporting information prevent a reliable assessment.

AIS coverage adequacy will be assessed using the frequency and continuity of available reports around the acquisition time, evidence of reception gaps, and supporting monitoring records where available. Consistency between estimated SAR target length and reported vessel dimensions may provide supplementary evidence where both measurements are sufficiently reliable. Anchorage location and proximity to piers will inform the review but will not automatically establish a match or explain an unmatched detection. The criteria for assessing coverage and assigning labels will be documented before model evaluation.

Initial agreement between reviewers will be reported using percentage agreement and Cohen's kappa. Disagreements will be resolved through joint review, while cases that remain unresolved will retain the Ambiguous/Insufficient Evidence label. The evidence source and confidence level for each reference label will be recorded. Independently verified cases will be distinguished from those supported only by reviewer assessment. Existing GFW classifications may provide supplementary information but will not be treated as independent ground truth.

The resulting reference labels will support the evaluation procedures described in Evaluation and Testing. Independently verified cases will form the primary evaluation subset where sufficient examples are available, while cases supported only by reviewer assessment will be analyzed separately. Ambiguous cases will be excluded from metrics requiring definitive labels, and their frequency will be reported. Synthetic AIS-dropout experiments will also be evaluated separately because simulated missing observations do not establish actual intentional AIS disabling. If sufficient independently verified cases cannot be obtained, the findings will be presented as preliminary results without claiming independently validated dark-vessel detection performance.

Framework: Heterogeneous Graph Construction and Modeling

The proposed framework will represent AIS trajectories and SAR vessel detections as a heterogeneous spatiotemporal graph for Batangas Bay. It will combine candidate vessel associations, temporal information, and local harbor context to assess cross-modal correspondence and identify unresolved discrepancies. The approach integrates link prediction for vessel association with node classification for discrepancy assessment.

1. Multi-Modal Node Formulation:

Each Sentinel-1 overpass will be represented by a graph with two node types: AIS vessel states and SAR detections. The design draws on graph-based approaches to vessel trajectories and spatial interactions (Wang et al., 2024; Zhang et al., 2023). Its basic structure is:

$$G = (V, E)$$

Here, G denotes the overpass graph, V contains both node types, and E contains the candidate and neighborhood connections between them.

For AIS nodes, vessel positions must first be aligned with the SAR acquisition time. Cubic Hermite spline interpolation will estimate these positions from sufficient observations within an initial 15-minute window before and after the overpass. Position and available velocity information will guide interpolation in a local metric coordinate system, with checks for implausible displacement or abrupt changes in motion. Each node will carry position, speed over ground, course over ground, and broadcast freshness. Freshness refers to the time separating the overpass from the nearest supporting raw AIS observation. Sparse or inconsistent

trajectories will retain uncertainty indicators because interpolation alone cannot guarantee positional accuracy.

SAR nodes will describe detected targets through geographic position, calibrated backscatter, distance from shore, and anchorage membership. Estimated target length will be retained for later discrepancy assessment. Keeping separate input features for AIS and SAR allows each modality to contribute its available evidence without assuming that both sensors measure the same vessel characteristics.

Before training, continuous features will be standardized using statistics fitted only on the training partition. Sine and cosine components will encode course so that headings on either side of north remain close in the feature representation. Binary indicators will record anchorage membership and missing features. Vessel identifiers will be used to assemble trajectories and enforce data partitions; excluding them from predictive inputs will limit identity-based memorization.

2. Graph Edge Structure and Physical Threshold Justifications:

Candidate edges will connect each SAR detection to plausible AIS nodes. A second group of edges will connect nearby nodes of the same modality, allowing the model to consider surrounding traffic. Both groups will use a distance-based gating rule:

$$d(i, j) <= r_{link}$$

In this rule, d(i, j) is the geographic distance between two nodes, and r_link is the search radius for the relevant connection type.

The initial search radii will be 5.0 km for candidate associations and 2.0 km for same-modality neighborhoods. Sensitivity analysis will determine whether these settings retain credible matches without introducing excessive competing candidates, taking AIS reporting gaps and local traffic density into account. A spatial KD-tree will retrieve neighbors from projected coordinates. Multiple candidates will remain available for graph inference, allowing the model to compare their evidence before selecting a correspondence.

Each candidate edge will carry spatial separation and AIS broadcast freshness. Heading and speed differences will be added where reliable estimates exist for both modalities, with angular comparisons accounting for course wraparound. Availability indicators will distinguish missing estimates from measured zero differences. Same-modality edges will supply context about vessel spacing and traffic density without implying a known AIS-SAR match.

These connections define three relation types: AIS-to-SAR candidate links, AIS-to-AIS neighborhood links, and SAR-to-SAR neighborhood links. Reverse candidate connections will permit message passing in both directions. The candidate subgraph is therefore bipartite, while the complete heterogeneous graph also contains spatial relationships within each modality.

3. Computational Architecture and Multi-Task Objective:

AIS and SAR features will enter separate linear projection layers before being mapped into a shared 128-dimensional embedding space with layer normalization. A two-layer Graph Attention Network using GATv2 and four attention heads will then pass messages across the graph (Brody et al., 2022). Relation-specific modules will handle candidate associations separately from same-modality traffic context. For each connection, attention will depend on the source and target embeddings together with available edge attributes. Normalizing these weights over the relevant neighbors will allow each node to combine messages according to their learned relevance.

After message passing, the final AIS and SAR embeddings will be concatenated with their candidate-edge attributes and passed to a multi-layer perceptron. A sigmoid output will produce an estimated match probability for each pair. These scores still require conflict checks before a unique correspondence can be accepted. A separate classification head will estimate discrepancy likelihood for each SAR node. The two tasks will be trained jointly through:

$$L_{total} = L_{association} + w * L_{classification}$$

Here, L_{total} is the combined loss. $L_{association}$ measures errors in candidate matching, $L_{classification}$ measures errors in SAR-node classification, and w controls the contribution of the classification task.

Candidate matching will use focal loss to reduce the influence of easily classified examples when non-matching pairs dominate the training data (Lin et al., 2017). SAR-node classification will use binary cross-entropy. Ambiguous or insufficient-evidence reference labels will be masked from the relevant supervised loss, although their nodes may remain in the graph as context. Dropout and validation-based early stopping will be evaluated to control overfitting. Loss weights, regularization, and focal-loss settings

will be selected under the time-blocked, vessel-disjoint validation protocol. The final test partition will remain reserved for evaluation.

4. Computational Resource Feasibility

Computational feasibility will be assessed on available academic hardware. During association, the model will process extracted vessel features, which reduces the input volume compared with full-resolution satellite imagery. A PyTorch Geometric HeteroData representation will store the typed node features, relation-specific edge indices, and edge attributes. Radius-based searches will keep connections local and avoid constructing every possible node pair. Benchmarks will record the trainable parameter count, saved model size, peak memory use, and training duration. Per-scene association latency will be measured as:

$$T_{scene} = T_{align} + T_{graph} + T_{model} + T_{filter}$$

T_{scene} denotes the total time to process one overpass. Its components cover trajectory alignment, graph construction, model inference, and contextual filtering.

Timing runs will cover different graph sizes and traffic densities, with hardware, batch settings, and node and edge counts documented. Measurements will begin after model loading and warm-up; GPU operations will be synchronized where applicable so that timings include completed computation. Repeated runs will establish median latency and its variation. Data download and raw SAR detection will be reported separately because they fall outside this feature-based association stage. Resource requirements and convergence times will be established through implementation benchmarks.

5. Anchorage-Aware Decision Logic

The final assessment will combine model outputs with anchorage and port-boundary polygons prepared in QGIS. Point-in-polygon queries will identify detections inside anchorages, while projected distances will indicate proximity to piers and coastal infrastructure. For each SAR detection, the strongest candidate association will be obtained through:

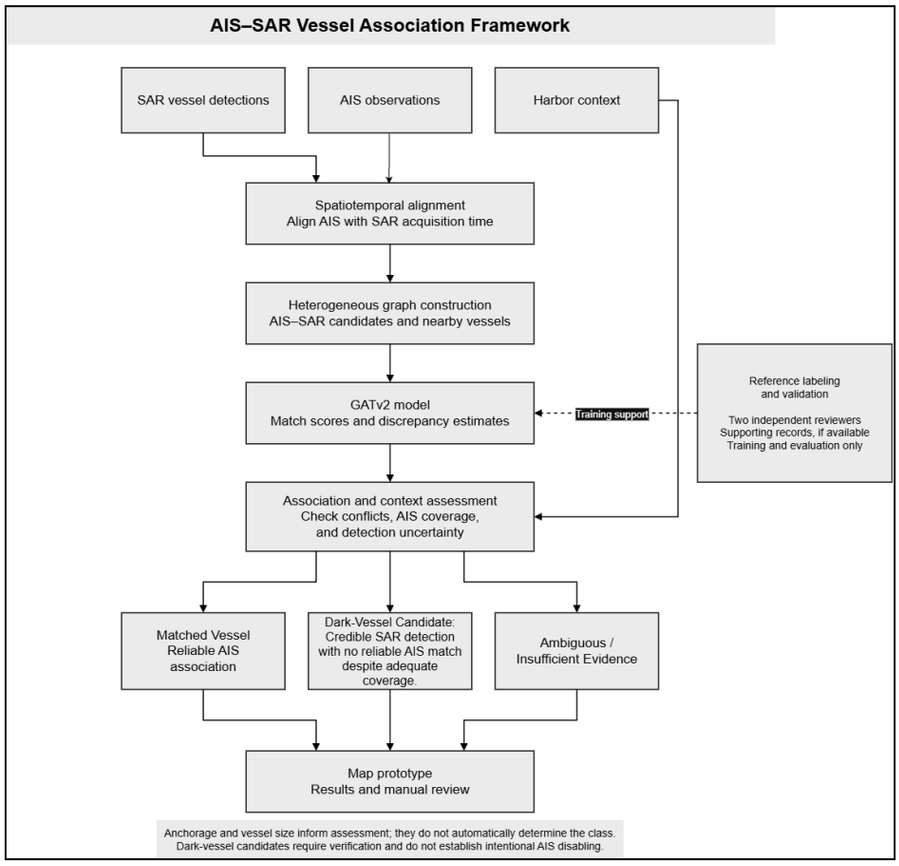
$$p_{best(i)} = max_j\, p_{_match(i,j)}$$

Here, $p_{best(i)}$ is the highest predicted match probability for SAR detection i across its candidate AIS nodes j.

An initial match threshold of 0.50 will be assessed over a validation range of 0.30 to 0.70. Similar scores among competing candidates will trigger an ambiguity flag, as will cases where several SAR detections favor the same AIS vessel. Detections with no candidate edges will be recorded as unassociated, followed by a check of AIS coverage adequacy. The selected threshold, remaining association conflicts, SAR-node discrepancy output, and harbor context will guide three output categories:

- Class 0 - Matched Vessel: The detection has a sufficiently confident AIS correspondence with no unresolved association conflict. This label records a match; it does not establish regulatory compliance.
- Class 1 - Dark-Vessel Candidate: A credible SAR detection has no reliable AIS correspondence despite adequate observation coverage, and the available context does not sufficiently explain the discrepancy. Such cases will be prioritized for manual review.
- Class 2 - Ambiguous / Insufficient Evidence: Competing matches or incomplete observations prevent a reliable assessment. Uncertain target dimensions and possible coastal clutter may also place a detection in this category. Anchorage location and slow movement will inform the review, but neither will automatically resolve an unmatched detection.

The proposed 25 m target-length threshold and 150 m pier buffer will be tested as contextual screening settings. They will not be treated as proof of AIS carriage obligations or intentional disabling. Consistent with the reference-labeling protocol, the framework's final output will identify unresolved observations that require further verification.



Concept Diagram

Benchmarking and Model Comparison

The proposed framework will be compared with baseline methods using the same held-out Batangas Bay observations and reference-labeling protocol. Vessel association and discrepancy assessment will be evaluated separately. All methods will use the same candidate-generation rules and available observations to support a fair comparison.

1. Baseline Methods

Two baseline methods will be used for vessel association:

- Distance-based Hungarian matching will assign AIS–SAR correspondences by minimizing spatial separation between SAR detections and AIS positions estimated at the acquisition time. Unmatched assignments will be permitted so that detections are not forced into unreliable correspondences.
- A pairwise multilayer perceptron (MLP) will estimate match probabilities from the available features of each AIS–SAR candidate pair without exchanging information with neighboring vessels. Comparison with this baseline will assess whether graph-based relational modeling improves association performance.

For discrepancy assessment, baseline association outputs will be combined with the same AIS coverage checks, ambiguity rules, and contextual assessment used by the proposed framework. This comparison will help distinguish improvements associated with the learned model from those provided by shared decision rules. Thresholds will be selected using validation data, while the test partition will remain reserved for final evaluation.

2. Ablation Experiments

Five configurations will be evaluated to examine the contributions of the proposed components:

- Full framework: The complete heterogeneous GATv2 model with candidate links, same-modality neighborhood links, edge attributes, and anchorage-aware context.
- Graph-free configuration: The pairwise MLP baseline without graph message passing.
- Anchorage-free configuration: The framework without anchorage-membership features or anchorage-based decision context.
- Neighborhood-free configuration: The framework without AIS-to-AIS and SAR-to-SAR neighborhood links.

- Edge-attribute-free configuration: The framework without edge attributes, while retaining node features and graph connectivity.

Configurations will use the same data partitions and comparable tuning procedures. Other settings will be kept consistent where applicable, and results will be compared across repeated runs to assess variation.

3. Evaluation Metrics

Vessel association will be evaluated using precision, recall, F1-score, and the area under the precision–recall curve (PR-AUC). Candidate-pair scoring metrics and final association results after conflict resolution will be reported separately.

An initial match threshold of 0.50 will be compared with thresholds from 0.30 to 0.70 using validation data. The selected threshold will be fixed before test evaluation.

Discrepancy assessment will be evaluated using precision, recall, F1-score, and false-positive rate. For this assessment, reference-labeled Matched Vessel cases will serve as negative cases, while Unresolved Discrepancy Candidate cases will serve as positive cases. A false positive will occur when a reference-labeled matched vessel is incorrectly flagged as a dark-vessel candidate. The false-positive rate will be calculated as FPR = FP / (FP + TN).

Cases labeled Ambiguous/Insufficient Evidence will be excluded from metrics requiring definitive reference labels. The proportion of detections receiving an ambiguous output will also be reported so that reduced false-alarm rates can be interpreted alongside the framework's ability to reach a decision.

Evaluation and Testing

The evaluation period and dataset partitions will be finalized after assessing overlapping AIS and SAR records. Chronological and vessel-disjoint partitions will separate training, validation, and testing. Feature standardization will use training data only, while model settings and decision thresholds will be selected using validation data before final testing.

Performance will be assessed using the reference-labeling protocol and metrics described above. Independently verified and reviewer-assessed cases will be reported separately. Controlled AIS-dropout experiments will examine sensitivity to incomplete reporting, with simulated conditions evaluated separately from observed discrepancies.

Where sufficient data are available, performance will be compared across traffic-density groups and anchorage conditions. Ablation experiments will assess the contributions of graph relationships, edge attributes, and anchorage-aware context.

Proof of Concept

A lightweight, interactive web-based map prototype will demonstrate the framework's vessel-association and discrepancy-assessment outputs for Batangas Bay. Recorded scenes will be used where suitable historical AIS and SAR observations are available. A live or periodically updated demonstration will be considered only if continuing access to sufficiently timely data can be secured.

The map will display SAR detections, available AIS trajectories, candidate associations, and anchorage boundaries where suitable geospatial records are available. Selecting a detection will show candidate matches, estimated match probabilities, spatial separation, time offsets, and relevant uncertainty indicators.

Results will be presented using three categories:

- Class 0 — Matched Vessel: A sufficiently confident AIS correspondence exists without unresolved association conflicts.
- Class 1 — Dark-Vessel Candidate: A credible SAR detection has no reliable AIS correspondence despite adequate coverage, and available context does not adequately explain the discrepancy.
- Class 2 — Ambiguous/Insufficient Evidence: Competing matches, incomplete coverage, uncertain detections, or insufficient supporting information prevent a reliable assessment.

Classification will follow the validation-selected threshold and the coverage, conflict, and contextual checks described in the methodology. Anchorage membership, estimated vessel length, and proximity to piers will inform the assessment without automatically determining the category.

The prototype will present the available evidence and reasons for further review of unresolved cases. It will support manual assessment, while a dark-vessel-candidate label will indicate a need for verification rather than establish intentional AIS disabling or a regulatory violation.

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November 10, 2026

MS. AURORA A. MENDOZA
Port Manager
Philippine Ports Authority – Port Management Office of Batangas
PPA Administration Building, Port Access Road
Calicanto, Batangas City, Batangas 4200

Subject: Request for Historical AIS Data and Supporting Vessel Records for Undergraduate Thesis

Dear Port Manager Mendoza:

Warm greetings from Batangas State University – The National Engineering University!

We are fourth-year Bachelor of Science in Computer Science students from the College of Informatics and Computing Sciences (CICS), Alangilan Campus, currently conducting our undergraduate thesis entitled **"Heterogeneous Spatiotemporal Graph Neural Network for AIS–SAR Vessel Association and Dark-Vessel Candidate Detection."**

Our study aims to develop a computational framework that integrates Automatic Identification System (AIS) vessel trajectories and Synthetic Aperture Radar (SAR) satellite observations to improve vessel association and identify potential maritime monitoring discrepancies in Batangas Bay.

In this regard, we respectfully request your office's assistance in accessing the following historical records, where available:

1. **Historical AIS Data.** Vessel position records, including timestamps, vessel identifiers, coordinates, speed, course, and available vessel characteristics, within ±30 minutes of 23 identified satellite acquisition times in 2025. The requested area covers latitudes 13.61°–13.83° N and longitudes 120.91°–121.08° E.
2. **Supporting Vessel Records.** Available vessel movement, berth or anchorage logs, and VTMS monitoring records that may assist in validating vessel associations.

To facilitate the request, we can provide the complete schedule of acquisition windows and initially request a sample for January 18, 2025, from 21:17 to 22:17 UTC, before proceeding with the remaining records.

We assure your office that all information obtained will be used exclusively for academic purposes and handled in accordance with applicable data privacy, confidentiality, and institutional policies. We are also willing to comply with any necessary documentation or data-sharing requirements.

Your assistance would be valuable in establishing the data feasibility and validity of our research. We sincerely hope for your favorable consideration.

Respectfully yours,

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Endorsed by:

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October 28, 2026

ATTY. MAXIMO I. BAÑARES, JR.
Director II
Maritime Industry Authority (MARINA)
Regional Office IV
Hinch Building II, Apacible Street, Barangay 10
Batangas City, Batangas 4200

Subject: Request for Technical Consultation and Academic Data Support for Undergraduate Thesis

Dear Director Bañares:

Warm greetings from Batangas State University – The National Engineering University!

We are fourth-year Bachelor of Science in Computer Science students from the College of Informatics and Computing Sciences (CICS), Alangilan Campus, currently conducting our undergraduate thesis entitled **"Heterogeneous Spatiotemporal Graph Neural Network for AIS–SAR Vessel Association and Dark-Vessel Candidate Detection."**

Our study aims to develop a computational framework that integrates Automatic Identification System (AIS) vessel trajectories and Synthetic Aperture Radar (SAR) satellite observations to improve vessel association and identify potential maritime monitoring discrepancies in Batangas Bay.

In this regard, we respectfully request the assistance of your office in the following areas:

1. **Technical Consultation.** An opportunity to interview appropriate MARINA personnel regarding AIS regulations, vessel identification, maritime monitoring challenges, and the practical relevance of our proposed research.
2. **Academic Data Support.** Guidance on the possible availability of non-confidential vessel information, including vessel classification, identifiers, dimensions, and other relevant records that may support the development and validation of our proposed model.

We assure your office that all information obtained will be used exclusively for academic purposes and handled in accordance with applicable data privacy, confidentiality, and institutional policies.

We would appreciate the opportunity to coordinate with your designated personnel at a mutually convenient schedule. Your technical guidance and expertise will be valuable in strengthening the practical relevance and validity of our research.

We sincerely hope for your favorable consideration.

Respectfully yours,

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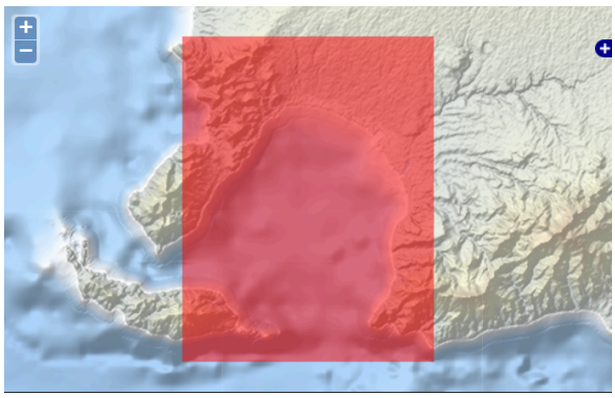
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Endorsed by:

DR. PRINCESS MARIE B. MELO
Dean, College of Informatics and Computing Sciences
Batangas State University – The National Engineering University

Study Area

Batangas Bay



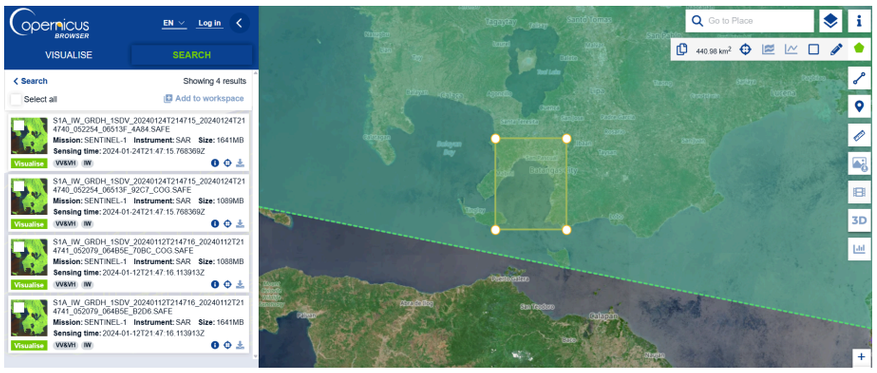
Latitude: 13.61° N to 13.83

Longitude: 121° 0' 2" E (121.00056°)

https://www.marineregions.org/gazetteer.php?p=details&id=17389

Dataset Sources and Feasibility Evaluation

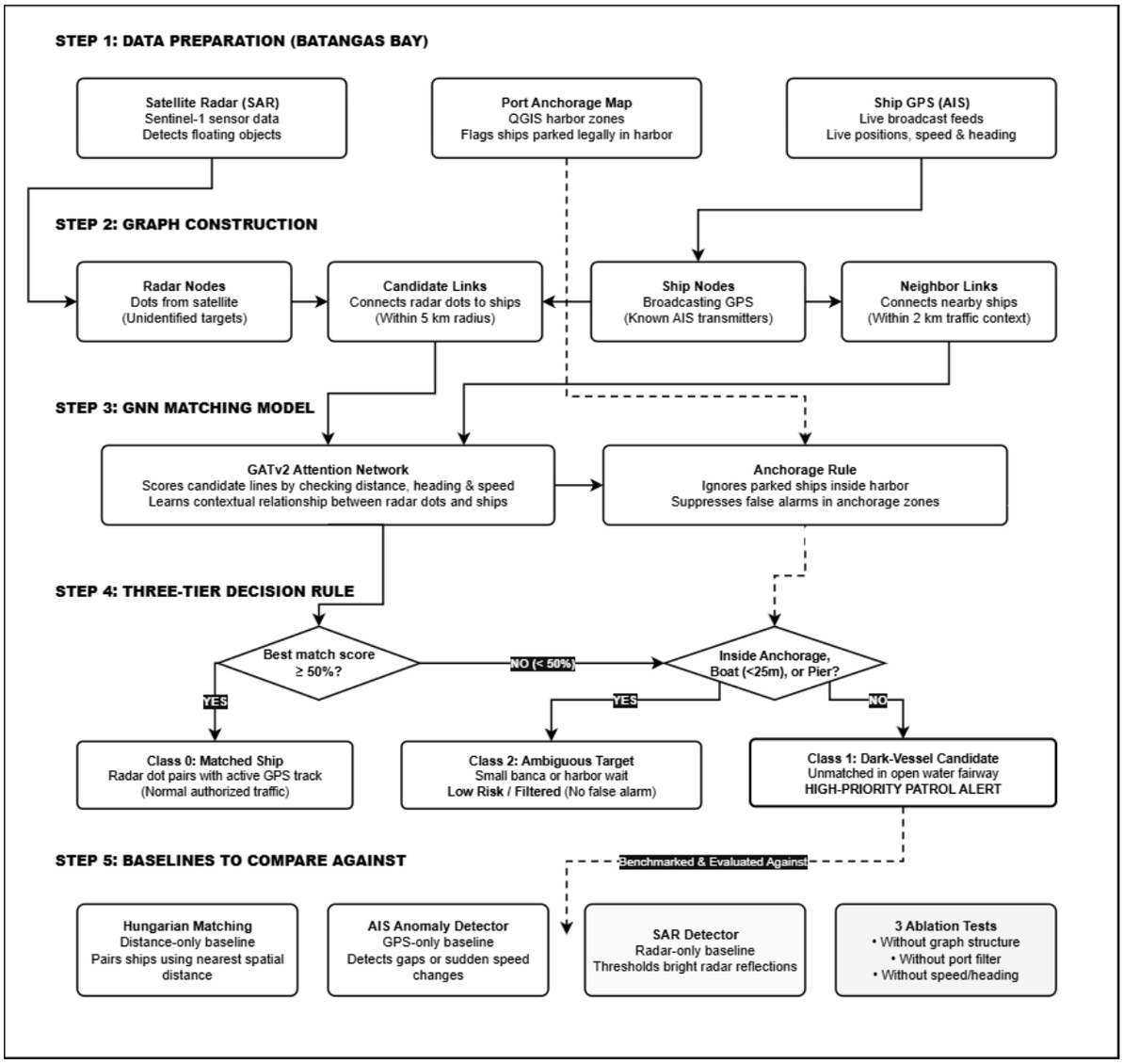
SAR: Copernicus Data Space Ecosystem



https://browser.dataspace.copernicus.eu/

A spatial-temporal query over the Batangas Bay bounding polygon on the Copernicus Data Space Ecosystem confirmed that Sentinel-1 captures the area at a consistent 12-day repeat cycle. For example, sample acquisitions in January 2024 occurred on January 12 and January 24 at approximately 21:47 UTC (05:47 AM PST). Across a full year, this 12-day cadence yields approximately 30 discrete cross-modal temporal epochs. Given the high vessel traffic density in Batangas Port, this provides an adequate sample of vessel detection instances to construct and benchmark the proposed cross-modal spatiotemporal graph.

Global Fishing Watch



1. Geographic Scope and Operational Cadence

The physical study area is confined to Batangas Bay (13.6◦–13.83◦N, 120.91◦–121.08◦E), bounding key industrial infrastructure, passenger ferry berths, commercial anchorage waiting zones, and the fairway corridor connecting toward the Verde Island Passage. Spaceborne observation relies on Copernicus Sentinel-1 Synthetic Aperture Radar (SAR), which revisits the bay every 12 days at approximately 05:47 AM Philippine Standard Time, yielding roughly thirty usable annual overpass scenes. While earlier sections outline the operational challenge of dark-vessel evasion, Sections 4 and 5 specify the computational pipeline: reconciling instantaneous satellite radar snapshots with asynchronous vessel radio broadcasts using a graph architecture that suppresses false alarms in congested waterways.

2. Key Architectural Decisions and Engineering Rationales

A. Selection of the 128-Dimensional Embedding Space

AIS transponders and spaceborne SAR capture fundamentally divergent modalities: kinematic motion fields (speed and course over ground) cannot be correlated directly with physical radar backscatter and vessel dimensions. To bridge this divide, independent linear projection layers map both data streams into a shared 128-dimensional latent space. This dimensionality strikes an intentional engineering balance: narrower latent spaces (16 or 32 dimensions) bottleneck complex interactions among velocity, orientation, and hull geometry, while excessively wide projections (512 or 1024 dimensions) trigger severe overfitting on compact harbor graphs containing fewer than one hundred simultaneous vessels. A 128-dimensional representation accommodates multi-head attention heads specializing independently in kinematic alignment, dimensional consistency, and geospatial context, while remaining lightweight enough to converge rapidly on consumer graphics hardware.

B. The 5.0-Kilometer Reachability Gating Threshold

Because satellite radar records an instantaneous snapshot while ship transponders broadcast intermittently, candidate broadcasts are assembled within a ±15-minute window (30 minutes total). The search radius around each radar detection is bounded by harbor vessel kinematics: commercial vessels navigating Batangas fairways operate under port speed limits between 8 and 12 knots, yielding a maximum physical displacement of roughly 5.5 km over a 15-minute interval. Setting a tighter gating threshold (under 3.0 km) prematurely drops legitimate vessels experiencing routine coastal VHF transmission latency, whereas expanding beyond 5.0 km causes spline interpolation to diverge and introduces dozens of spurious pairings. The 5.0 km threshold establishes a physically grounded reachability envelope, connecting each radar detection to five to twelve candidate vessels for efficient attention-based scoring.

C. Geospatial Harbor Zoning via QGIS

Raw radar sensors cannot differentiate between administrative maritime zones; an authorized vessel waiting at anchor exhibits the same radar backscatter as an unbroadcasting vessel loitering in open shipping lanes, while concrete piers scatter radar energy intensely. To inject operational context, official NAMRIA hydrographic charts and Philippine Ports Authority port layouts are digitized in QGIS to produce vector polygons for designated anchorages, navigation fairways, and a 150-meter shoreline buffer. During pipeline execution, GeoPandas queries each detection against these geometries: detections inside anchorage grounds receive dedicated context indicators, while land returns and near-pier reflections are pruned as coastal clutter. This geospatial feature informs the graph network that stationary anchored vessels represent compliant waiting rather than illicit activity.

D. The Three-Tier Labeling Protocol

Treating every unbroadcasting radar detection as an illicit dark vessel introduces severe labeling bias. Under IMO SOLAS regulations, artisanal fishing boats, small tugs, and motorized craft under 25 meters are legally exempt from transmitting AIS, while shoreline infrastructure routinely mimics floating targets. The framework resolves this through a deterministic three-tier decision rule: Class 0 designates AIS-matched compliant vessels; Class 1 flags high-priority dark-vessel candidates (unmatched radar targets over 25 meters actively navigating open fairways outside anchorages); and Class 2 isolates ambiguous targets, including sub-25-meter artisanal craft, anchored vessels, and near-shore scatter. Masking Class 2 during loss calculation prevents statutory exemptions and coastal noise from corrupting optimization gradients, directly driving down false alarms.

E. Dynamic Graph Construction and Pipeline Flow

Rather than enforcing greedy nearest-neighbor pairings that trigger cascading identity swaps across crowded queues, the architecture formulates vessel correspondence as dynamic link prediction over a heterogeneous graph. Trajectory broadcasts within 15 minutes are interpolated to the overpass second using cubic splines, forming AIS nodes populated with kinematic features, while radar targets form SAR nodes with spatial, geometric, and zoning attributes. A spatial KD-tree connects each radar target to candidate AIS nodes within 5.0 km via reachability edges storing spatial separation, broadcast latency, heading deviation, and velocity feasibility. Intra-modal neighbor edges link vessels within 2.0 km of one another, supplying the network with vital local queue awareness.

Packaged into a PyTorch Geometric HeteroData structure, a two-layer Graph Attention Network (GATv2) passes contextual messages across candidate and neighbor links simultaneously. The attention mechanism dynamically computes connection weights across all potential pairings, outputting continuous match probabilities. Radar detections whose candidate edges all fail to exceed the 50% match threshold are flagged as unassociated targets and evaluated against the anchorage filter and 25-meter threshold to confirm actionable dark-vessel candidates.

3. Open-Source Data Acquisition Roadmap & Source Comparison

The framework is fully implementable using publicly accessible, non-commercial maritime data across complementary spaceborne and terrestrial sources:

Synthetic Aperture Radar (SAR): Primary radar coverage is sourced from the Copernicus Data Space Ecosystem, which supplies raw Sentinel-1 C-band SAR scenes on a 12-day repeat orbit for baseline verification. To eliminate heavy raster preprocessing, analysis-ready vessel detections are obtained via Global Fishing Watch (public-global-sar-presence), providing calibrated radar backscatter, bounding lengths, and coordinates. Local downlinked passes and high-performance computing support are mirrored through the DOST-ASTI PEDRO Center and Project DATOS archives.

Vessel Trajectory Telemetry (AIS): Historical 2025 AIS trajectories are acquired from Global Fishing Watch's open data repository, providing pre-synchronized kinematic tracks without requiring live streaming APIs. Local ground-truth validation is complemented by shore-based radar and transponder logs requested from the Port of Batangas Vessel Traffic Management System (PPA PMO Batangas), with national surveillance feeds supported through DOST-ASTI's SIYASAT platform.

Nautical Boundaries and Global Pre-Training: Geographic fairway corridors and anchorage polygons are digitized from OpenSeaMap and verified against official NAMRIA nautical charts (PH4BTS40 and 4256A). To eliminate risk from localized data sparsity, the architecture incorporates the open xView3-SAR benchmark dataset (>1,000 Sentinel-1 scenes with paired AIS ground truth) to pre-train the GATv2 backbone before fine-tuning on Batangas Bay observations.